LLMs vs Google: The Future of Information Retrieval

A positional Paper
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Abstract

Over the past few years, large language models (LLMs) such as ChatGPT, Claude, and Gemini have changed the way people search for information. Unlike traditional search engines like Google, which mainly depend on keyword matching and ranking systems, LLMs are able to understand context and provide answers in a more conversational and meaningful way and short explanations give users less time to spend on any query .

In this paper, I argue that LLMs offer more sufficient, clear, and user-focused responses compared to Google Search. This shows that there is a shift in how digital information is retrieved and understood.

1.Introduction

The way people search for information has changed a lot over time with the growth of technology. Earlier, people used books and encyclopedias. Then the internet came, and tools like Yahoo and Google became popular for searching information. Each stage brought a new way of finding answers. Today, we are seeing another big change with the rise of Large Language Models (LLMs). These are AI systems trained on large amounts of data that can understand and generate human-like text. Many people now use them daily not just for writing, but also for searching information.

Traditional search engines like Google mainly work by matching keywords and showing a list of links. Users then need to open different websites to find their answers. On the other hand, LLMs give direct answers in a conversational way. They combine information and present it in a clear and structured format. This makes it easier for users to understand topics quickly without going through many links.

In this paper, I argue that LLMs provide more clear, fast, and user-friendly responses compared to Google Search for many everyday queries. This shows that the way people access and understand information is changing. However, based on my own observations, LLMs also have some limitations. They are not always reliable when it comes to real-time information, verifying sources, or doing deep research from detailed articles and sometimes

hallucinations. Because of this, both LLMs and Google Search have their own strengths, and they can be used together depending on the situation.

2.Hypothesis

Before starting the experiment, the following hypothesis was made-

LLMs will provide faster, more contextual, and more user-friendly answers compared to Google for understanding concepts, building intuition, and getting structured explanations. However, for tasks that require deep research, detailed articles, real-time information, or fact verification, users may need to switch to Google. This suggests that LLMs are strong tools for support, but they cannot fully replace traditional search engines.

This hypothesis is based on two main observations. First, LLMs can answer questions in natural language, so users do not need to go through multiple links to understand a topic. Second, LLMs may sometimes give incorrect or incomplete information because they are trained on past data and the same hallucination trap and do not always have real-time updates. Because of this, checking information from other sources becomes important, especially for important or sensitive queries.

3.Methodology

To test the hypothesis, a structured personal experiment was conducted over five days, from April 10 to April 14, 2026. In this experiment, four LLMs — ChatGPT, Claude, Gemini, and Perplexity — were used as the main search tools for both academic and general queries.

During the experiment, a log was maintained to record the following details:

- Date of the query
- LLM used
- Google query (if there was a need to switch to Google)
- The prompt given to the LLM
- Observation: what worked, what did not work, and whether switching to Google was required

In total, 30 queries were recorded over the five-day period. ChatGPT and Claude were each used for 8 queries, while Gemini and Perplexity were each used for approximately 6 queries. The queries covered different topics such as academic writing, computer science, mathematics, automation, psychology, global labor markets, and game development and hallucination tests.

The collected data was later analyzed to compare the performance of each LLM and to understand when switching to Google was necessary. The distribution of queries across each LLM is shown in Figure 1

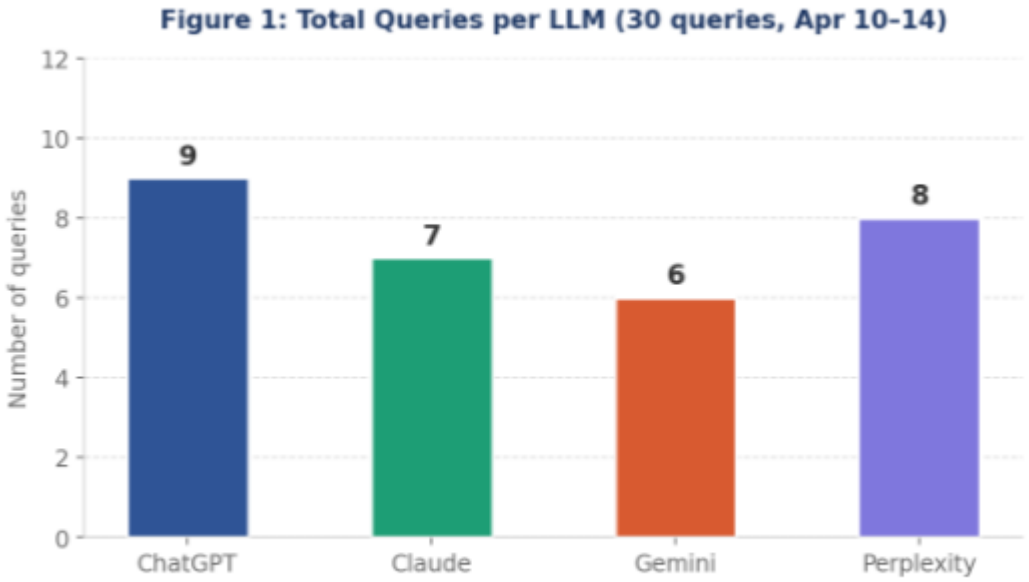


Figure 1: Total queries logged per LLM across all 30 entries (Apr 10–14, 2026)

4.Observations and Results

Out of the 30 total queries recorded, 21 queries were fully answered using only LLMs (about 70%), while 9 queries required switching to Google for additional information (about 30%). Figure 2 shows this overall success rate.

Figure 2: LLM Success Rate vs Google Switch Rate (30 total queries)



4.1 When LLMs Succeeded

Out of the 30 queries, 22 were answered successfully by the LLMs without needing to switch to Google. These successful cases had a common pattern: the user was looking for basic understanding, structured guidance, or a quick and simple explanation of a topic.

For example, when asked about how automation affects productivity, Perplexity gave a clear and well-structured answer, explaining both the positive and negative effects. There was no need to switch to Google in this case.

If a user wants to generate a question paper giving the pattern claude give it like nothing at one prompt which leads to not to depend on search engine like use case.

At the same time user requirement is only answers from a pdf there is no need to use google one sided for this type of use case can be covered.

Similarly, Gemini provided a clear answer when asked about the role of mathematics in computer science. It explained important topics like linear algebra, discrete mathematics, and probability in a simple and organized way. ChatGPT also performed well when asked to give a structured guide for writing a research paper, as it provided a step-by-step explanation that was easy to follow.

The main observation from these cases is that when the user needed quick understanding instead of reading multiple sources, LLMs were able to give clear and helpful answers efficiently

4.2 When Google Became Necessary

In this part, we observed that 9 out of 30 queries (30%) needed the user to switch to Google. By analysing these cases, we can understand some clear reasons for this behavior.

One important reason is trust. When Perplexity was asked whether LLMs can harm human thinking, the user was not satisfied with the answer. The user said that they needed to check Google for confirmation. This shows that when questions are about AI itself, users often do not fully trust LLM answers and prefer external sources.

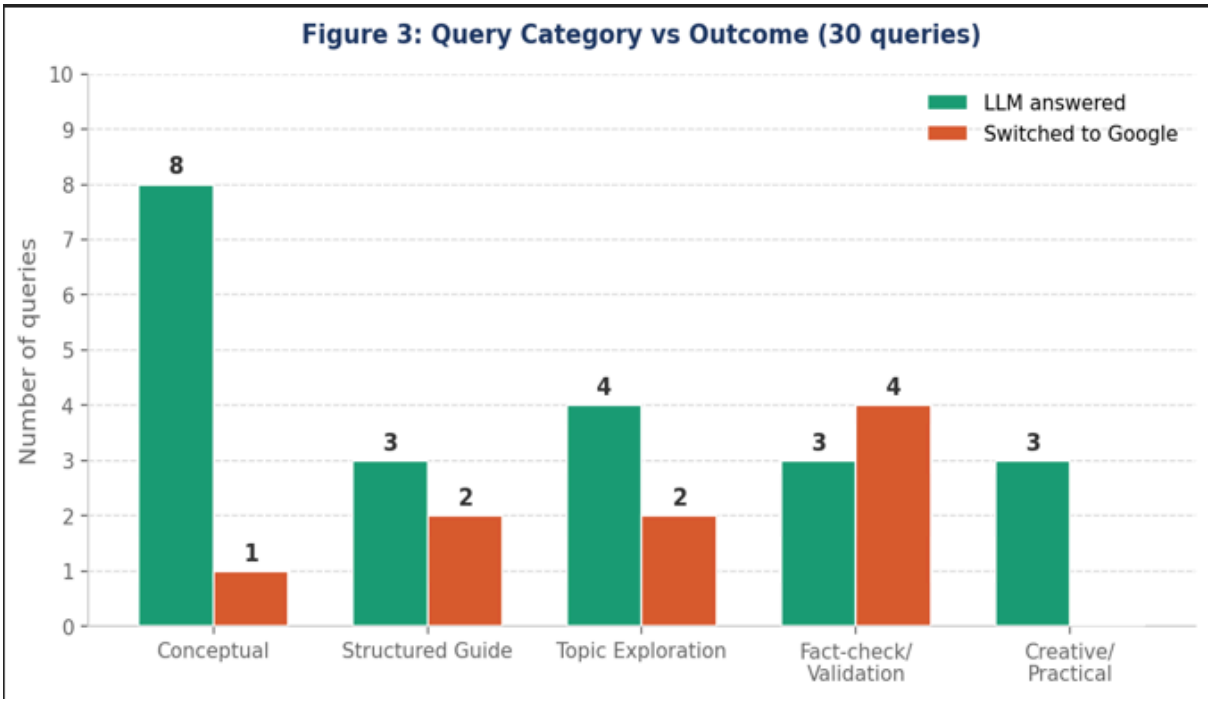
Political questions also showed limitations of LLMs. For example, when ChatGPT was asked about BJP vs TMC in 2026, it gave a prediction instead of real and current data. The user needed factual and updated information, which Google can provide through live news. In another case, Claude was asked about mobile phones under 25k, but the suggestions were not based on current market prices. This shows that LLMs may not always give real-time or accurate data.

One thing can be observed that when the query is more research oriented and the user has time to go through the question then Google got the score showing all variety of links so the user got his clear answer.

Google is helpful when Price and location is asked by the user then no LLMs is as close as google search.

Another query tested how LLMs respond to harmful or illegal requests, such as bomb-making instructions. Both Claude and Google refused to answer. This was noted as a test of safety limits. It shows that LLMs are now similar to Google in rejecting unsafe content.

Overall, Figure 3 shows these results by different types of queries. It helps to clearly understand when LLMs work well and when users need to depend on Google.

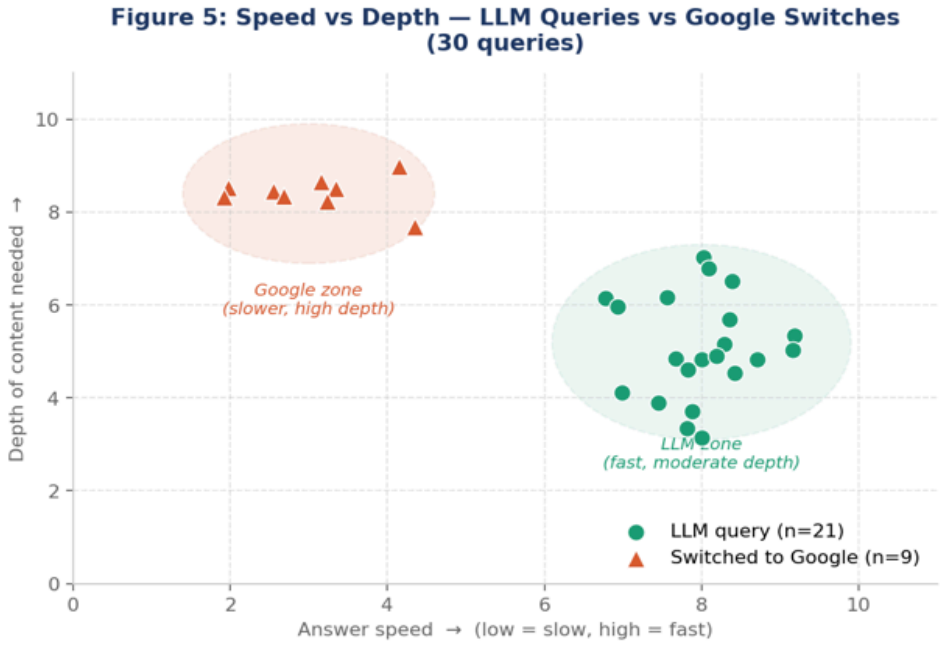


5. Analysis and Discussion

5.1 Speed vs Depth - The Core Trade - Off

The main finding of this study is that LLMs and Google serve different needs of users. LLMs are designed for speed and summarising information. They help users understand a topic quickly by giving a structured answer. On the other hand, Google is designed for depth and variety. It helps users find original sources, articles, research papers, and updated web pages.

Figure 5 shows this difference clearly. It plots each query based on speed and depth. LLM queries are shown as green circles, while cases where users switched to Google are shown as orange triangles. The pattern shows that LLM queries are mostly in the high-speed and medium-depth area. In contrast, Google-related queries are in the lower-speed but higher-depth area.



5.2 The Trust and Hallucination Problem

One important finding of this study is the role of trust. Users switched to Google not only when the LLM answer was clearly wrong, but also when they were unsure. Even if the answer looked correct, users still wanted to verify it. This shows a known problem of LLMs: they can give confident answers that are sometimes incorrect. This problem is called hallucination.

Hallucinations—the generation of factually incorrect information presented with high confidence—remain the primary obstacle to LLM reliability. Research from MIT in 2025 revealed that LLMs actually use more confident language, such as "definitely" or "without a doubt," when they are hallucinating than when they are stating facts. This creates a dangerous scenario for users who rely on LLMs for deep research without cross-referencing.

The log entry for this case is very meaningful. The user said that LLM is like a smart storyteller that can give wrong answers with confidence. This explains why Google is still important. Google shows different sources, so users can check and verify the information by themselves.

Until LLMs improve in giving verified and real-time information with proper citations, Google will continue to be more reliable for important and factual queries.

5.3 LLMs as Cognitive amplifiers

A key observation from using these tools daily is that LLMs act as cognitive helpers. They do not just give information, but also help users think better. A good LLM answer explains the reasoning, suggests different options, and helps users understand a topic step by step. And build a clear intuition of any topic in less time. This is something Google cannot do in the

same way, because it mainly provides links and expects users to read and organise the information themselves.

From the data, about 70% of the queries were in categories where LLMs performed this role successfully. These include building intuition, explaining concepts, giving structured guidance, and helping with code or content creation. This shows that for most everyday information needs, LLMs are very useful.

6.Conclusion

This study aimed to understand whether LLMs can provide a better search experience than Google and if they can replace traditional search engines. Based on five days of experiments with 30 queries across four LLMs (ChatGPT, Claude, Gemini, and Perplexity), the results give a clear and balanced answer.

LLMs performed better than Google for most everyday tasks. Around 70% of the queries were fully answered by LLMs without needing Google. For tasks like understanding concepts, getting structured explanations, writing code, and quick learning, LLMs were faster, easier to use, and more helpful.

However, LLMs cannot fully replace Google. About 30% of the queries still required Google, especially for real-time information, current events, market data, and fact-checking. These limitations come from how LLMs work, such as limited real-time data access and the possibility of incorrect answers (hallucination).

Google search engines are one of the foundational tools behind the development and usefulness of modern LLMs. Because of this, they remain important for verifying the results generated by LLMs.

So, the correct way to see this is not “LLMs vs Google”, but “LLMs and Google”. Both tools serve different purposes and work best together. LLMs are useful for understanding and summarising, while Google is better for verification and deep research.

In the future, information searching will likely use both systems together. LLMs will handle explanation and thinking support, while search engines will provide trusted sources and updated information. Over time, as LLMs improve, their role will continue to grow in this combined system.

References

1.Claude- https://claude.ai/new
2.Chatgpt- https://chatgpt.com/
3.Gemini- https://gemini.google.com/app
4.Perplexity- https://www.perplexity.ai/
5.Google- https://www.google.com/